

The *Notch* Locus of *Drosophila melanogaster*

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Summary

The *Notch* locus appears to correspond to a 37 kb transcription unit. Homologous poly(A)⁺ RNA is about 11.7 kb. Nine RNA coding regions have been detected and localized within the transcription unit by S1 nuclease analyses. These range in size from 130 to 7250 bp. Sequences within the 3' coding region are variably used. Twenty-four *Notch* locus "point" mutations have been examined. Seven are associated with DNA insertions. Four insertions are associated with dominant mutations and are located within or very near RNA coding sequences of the 37 kb transcription unit. Three insertions are associated with recessive mutations and fall within intervening sequences. The positions of all insertions agree with the genetic map. It is estimated that, in the *Notch* region, one map unit corresponds to approximately 275 kb. On this basis, most of the 17 mutations that were not associated with DNA insertions can be placed within coding sequences of the 37 kb transcription unit.

Introduction

Mutations at the *Notch* (*N*) locus fall into several classes that affect the morphology of the fly in apparently unrelated ways. One class of mutations alters the formation of the wings, while another affects the eyes and bristles. A third class is associated with lethal changes in the development of the embryonic nervous system (reviewed by Wright, 1970) and also decreases the levels of activity of several adult mitochondrial enzymes (Thorig et al., 1981). Mutations in this class are referred to as *N*, or *Notch*, mutations. They are lethal when homozygous and produce a dominant notched wing phenotype when heterozygous with a wild-type gene. *N* mutations are scattered throughout the *Notch* locus, spanning nearly 0.13 map units (Figure 1); this is 19 times the size of *rosy*, which is estimated to be about 4 kb (Spradling and Rubin, 1981). *N* mutations do not complement each other or any other mutations of the locus, and deletion of the entire *Notch* region gives a phenotype comparable to that of an *N* mutant. Cytogenetic analyses have placed *Notch* in chromomere 3C7 or its proximal interband (Slizynska, 1938; Welshons and Keppey, 1981).

Notch is one of many loci in *Drosophila* that appear to be complex when analyzed by intra-allelic complementa-

tion tests (reviewed by Judd, 1976; see also Lindsley and Grell, 1968). In addition to the *N* mutations, another class of dominant alleles, *Abruptex* (*Ax*), is found at the locus. These mutations produce a wing vein gapping phenotype when heterozygous with a wild-type gene. Some *Ax* alleles are viable and produce the wing phenotype when homozygous, while others are homozygous lethals (Welshons, 1971; Foster, 1975; Portin, 1975). From complementation analyses with *N* mutations, *Ax* mutations can be divided into two groups: those that enhance the *N* mutant phenotype and those that suppress it. Heterozygotes of different *Ax* alleles sometimes give negative complementation; the phenotypes of the heterozygotes are more aberrant than those of the respective homozygotes. All of the *Ax* mutations that have been located by recombination lie to the right of *spl* (see Figure 1).

There are several groups of recessive mutations within the locus (Welshons, 1965; Lindsley and Grell, 1968). Two of these groups are typically characterized by their effects on eyes, *facet* (*fa*) and *facet glossy* (*fa*^g), or on wings, *facet notchoïd* (*fa*^{no}), *notchoïd* (*nd*), and *notchoïd two* (*nd*²). In heterozygotes, the two eye mutations fail to complement each other, as do all heterozygous combinations of the three wing mutations. Eye and wing mutations complement each other and thus appear to be in different genes. Another recessive eye mutation, *split* (*spl*), complements the wing and eye mutations described above, so it seems to identify a third gene. Finally, several recessive lethals are found at the locus—*l(1)N*, *l(1)N*², *l(1)N*³, and *l(1)N*^B (Welshons, 1965; Wright, 1970). Some of these are similar in lethality to the *N* mutants (discussed below). In complementation tests with eye and wing mutations, one of them, *l(1)N*, behaves like a mutation in a fourth gene.

In spite of the apparent subdivision of these recessive mutations into separate genes shown by complementation tests, all must be part of a single functional unit because they fail to complement *N* mutations. Support for the conclusion that a functional unit encompasses both recessive and dominant mutations comes from genetic recombination analyses, which indicate that *Notch* mutations exist on both sides of many of the recessive and *Ax* mutations (Figure 1).

A variety of experiments have shown that *Notch* is active at several developmental stages. In homozygous *N* embryos, the ectodermally derived nervous system contains three times as many cells as normal. This enlargement is at the expense of another tissue of ectodermal origin, the hypoderm, and results in death late in embryogenesis. A similar phenotype is seen with the recessive lethal mutants *l(1)N*, *l(1)N*², and *l(1)N*³ (reviewed by Wright, 1970). Aberrant *Notch* locus expression during the second and third larval instars and pupation can be lethal or can be associated with viable mutant phenotypes (Shellenbarger and Mohler, 1975, 1978). In an attempt to understand the wide-ranging activity of the *Notch* locus and the complementation patterns of its mutations at a molecular level, we have

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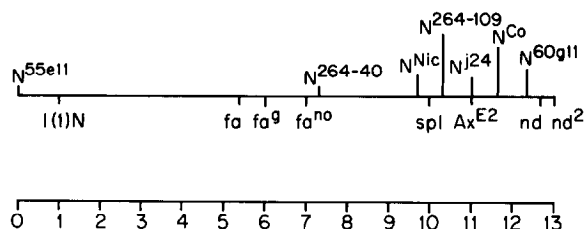


Figure 1. The Recombination Map of the *Notch* Locus

The positions of recessive mutations and a single *Abruptex* mutation are shown below the top line. Above it are indicated the positions of a selected sample of dominant *Notch* mutations. The lower line shows recombination distance (total length is 0.13 map units). With the exception of *l(1)N*, *N264-109*, and *Ax'E2*, the positions of mutations are taken from Welshons (1974). *Ax'E2* was located by Foster (1975), and *l(1)N* and *N264-109* by Welshons (personal communication).

isolated DNA corresponding to the gene, and we present here a molecular characterization of the *Notch* locus.

Results

Isolation of *Notch* Locus DNA

Sequences from the *Notch* locus were identified for cloning by use of the inversion *ln(1)N^{76b8}* (see also Artavanis-Tsakonas et al., 1983). Cytologically, this inversion is characterized by one breakpoint in the 3C7–3C8 region of the X chromosome and another at about 3C10 (Keppy and Welshons, 1980; see Bridges, 1935, for nomenclature). Genetically, the distal break of this inversion has been mapped within the *Notch* locus between the mutations *spl* and *N264-40* (Welshons and Keppy, 1981). A library of cloned Eco RI fragments was constructed from flies carrying this inversion (see Experimental Procedures). The proximal breakpoint was isolated by using sequences previously cloned from the 3C10 region as a hybridization probe (Muskavitch and Hogness, 1980). A restriction fragment from this clone was used to isolate a portion of the *Notch* locus from the wild-type Canton S DNA library of Maniatis et al. (1978). Overlapping segments of *Notch* DNA were isolated in both directions. A total of about 120 kb of DNA was obtained, and a restriction map of the relevant part of it is shown in Figure 2.

Under the hybridization conditions used (see Experimental Procedures), two repetitive sequences, named strep and flea, were found in this region. The strep sequence is between coordinates 7.8 and 8.3 and is approximately 500 bp or smaller in size. From the number of plaques with which it hybridizes in the Canton S library (8%), as compared to several unique sequence probes, it is calculated to be present approximately 400 times per haploid genome. Similar patterns of hybridization are found when strep is hybridized to restriction fragments from several sibling species of *D. melanogaster* (S. Kidd, unpublished result), suggesting that it is a stably located repetitive sequence. At about position +16 the Canton S library is polymorphic for the presence of a repetitive 5 kb copia-like element, flea (discussed below).

To confirm that the isolated DNA is from the *Notch*

locus, the breakpoints of three additional chromosomal rearrangements, *Df(1)N⁶⁶²⁵*, *Df(1)N^{62b1}*, and *Df(1)N^{68f19}* were located on the physical map (Figure 2b). The chromosome rearrangements produce a *Notch* phenotype, and their breakpoints have been mapped to discrete subregions of the locus by recombination studies (Welshons, 1974; Welshons and Keppy, 1981). The breakpoints were identified by hybridizing phages, according to the techniques of Southern (1975), to restriction-endonuclease-cleaved DNA from flies with the chromosome rearrangement and a duplication to cover its lethality. After a breakpoint was localized to a particular phage, gel-purified restriction fragments were hybridized to the DNA to position the breakpoint more accurately. These experiments are shown in Figure 3, and the location of each breakpoint is given in Figure 2 and its legend. All four of the chromosome rearrangement breakpoints, including *ln(1)N^{76b8}*, occurred in the same order and were separated by the same relative distances as had been predicted by genetic recombination mapping.

Transcription Mapping of the *Notch* Region

To identify regions of the cloned 120 kb of DNA homologous to RNA, phage clones were hybridized to electrophoretically fractionated poly(A)⁺ RNA from staged embryos, first, second, and third instar larvae, pupae, and adults. Several transcripts were found by this method in the 120 kb of cloned DNA (S. Kidd, unpublished result), but only those found in the region containing the *Notch* chromosome rearrangements are described below. Figure 4a shows the result of hybridizing lambda N2 (−8.4 to +6.2) to poly(A)⁺ RNAs from various stages of the *Drosophila* life cycle. A transcript of about 11.7 kb (see Figure 4 legend) is found that is present in all stages but is most abundant during embryogenesis, 8–12 hr after egg laying, and in pupae. A less abundant, smaller transcript of about 9 kb is seen in some preparations from 0–4 hr embryos (Figure 4a), and is currently being characterized. The same filter shown in Figure 4a was also hybridized with lambda NL22 (−22.2 to −8.2) and NL35 (−33.9 to −19.5). The former failed to detect any discrete transcripts, but the latter hybridized to an 11.7 kb transcript with the same developmental profile as that found with lambda N2 (data not shown). When a different filter was used, restriction fragments from lambda NR311 (+5.8 to +21.0) were also found to hybridize to an 11.7 kb RNA (data not shown).

The direction of transcription of the 11.7 kb RNA detected by these phages was determined by using strand-specific probes produced by cloning restriction fragments (a 4.9 kb Eco RI fragment, coordinates −30.7 to −25.8, from lambda NL35; a 1.75 kb Bam HI fragment, +2.7 to +4.5, from lambda N2; and a 1.1 kb Eco RI/Bgl II fragment, +9.0 to +10.1, from lambda NR311) into the single-stranded phage vectors M13mp8 and mp9. These were ³²P-labeled by the procedure of Hu and Messing (1982) and hybridized to electrophoretically fractionated poly(A)⁺ RNA. Figures 4b and 4c show the result of hybridizing the separate strands of the 1.75 kb Bam HI fragment. Only

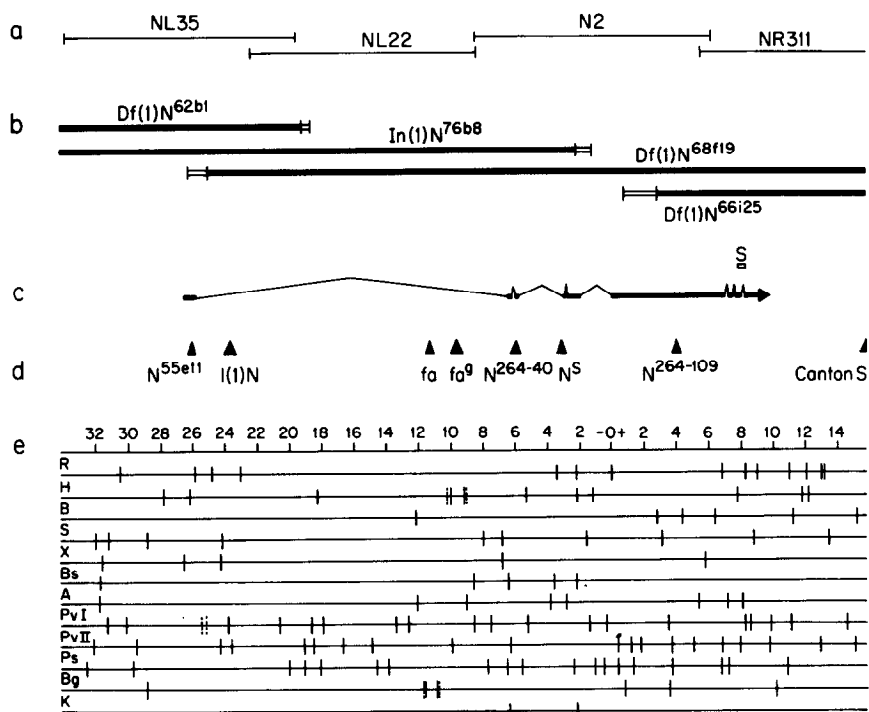


Figure 2. A Physical Map of the Notch Locus

(a) Locations of overlapping phages containing the locus. Additional phages were isolated in the -8.5 to $+20$ region, most of which contain a copia-like element, flea, at position $+15.3$ to $+16.6$. (b) The locations for four chromosome rearrangements that break within the locus. Bars represent DNA still present and in normal sequence in the rearrangement. Open areas of the bars represent region in which the breakpoint is located. These are -26.2 to -24.7 , $Df(1)N^{68f19}$; -19.3 to -18.6 , $Df(1)N^{62b1}$; -2.2 to -1.3 , $In(1)N^{76b8}$; $+0.8$ to $+2.7$, $Df(1)N^{66i25}$. (c) Location of coding regions homologous to the 11.7 kb RNA. The location of the third coding region is approximate (see text). Coding regions smaller than ~ 130 bp would probably have gone undetected in the mapping experiments. S indicates the location of the dispersed, repetitive sequence strep. Hybridization experiments with a non-Notch strep sequence show annealing only to the eighth coding region (S. Kidd, unpublished). Under low stringency ($0.5\times$ SSC at 65°C) washing conditions, strep hybridizes with many RNAs ranging in size from 1 to 12 kb. The majority of this hybridization is removed by washing in $0.1\times$ SSC at 65°C (S. Kidd, unpublished result). This sequence appears to have some of the properties of the "suffix" repeat sequence (Tchurikov et al., 1982). (d) Locations of insertions found on various chromosomes. With the exception of Canton S, which contains flea at the indicated position, the insertions are associated with mutant phenotypes (see text). The width of the base of each triangle indicates the region in which the insertion was located. (e) Restriction map and coordinate scale for the cloned Canton S DNA. The locations of Eco RI (R), Xho I (X), Hind III (H), Bam HI (B), Sst I (S), and Bgl II (Bg) sites were determined on phage clones. The rest, Bst E II (Bs), Ava I (A), Pvu I (PvI), Pvu II (PvII), Pst I (Ps), and Kpn I (K), were mapped on subcloned adjacent restriction fragments. Kpn I sites were only mapped within two adjacent Eco RI fragments of N2 (coordinates -8.4 to -2.2). In some cases, restriction fragments less than 400 bp would have gone undetected. Pairs of dashed lines indicate that a site is located at only one of the indicated positions. The coordinate scale is in kilobase pairs with zero being the first Eco RI site in Canton S DNA proximal to the breakpoint of $In(1)N^{76b8}$. Negative numbers go toward the telomere, and positive ones toward the centromere.

the m13B11 probe (Figure 4b), which would detect an RNA transcribed from left to right of Figure 2, hybridizes to the 11.7 kb transcript. The same direction of transcription was found with the other two probes (data not shown). This suggests that the 11.7 kb RNA is formed by the splicing of a precursor RNA transcribed from sequences in lambda NL35, NL22, N2, and NR311.

The sequences in these phages that are homologous to the 11.7 kb RNA were located approximately by hybridizing restriction fragments to electrophoretically fractionated poly(A)⁺ RNA. This defined a region of patchwork homology to the 11.7 kb RNA extending from the 1.5 kb Hind III fragment (-27.7 to -26.2) to the 1.1 kb Eco RI/Bgl II fragment ($+9.0$ to $+10.1$). No homology to an 11.7 kb RNA was found with DNA extending 48 kb to the left and 30 kb to the right of this region. The RNA homologous regions were then located more accurately by S1 nuclease mapping and were shown to be spliced by the two-

dimensional S1 mapping procedure of Favaloro et al. (1980). Several S1 mapping experiments are shown in Figure 5.

To map what appears to be the 5' coding region, pupal RNA was annealed with Bam HI-, Eco RI-, or Xho I-cleaved lambda NL35, treated with S1, fractionated through an alkali gel, then hybridized with a ³²P-labeled Xho I/Eco RI fragment (-26.5 to -25.8) that is homologous only to the 11.7 kb RNA. A 0.8 kb fragment is detected with either Bam HI- or Eco RI-cleaved lambda NL35; this is cleaved by Xho I, producing a 0.65 kb fragment (Figure 5a). This positions one end of the coding region 0.65 kb to the right of the Xho I site at -26.5 , and by subtraction the other end is 0.15 kb to its left (i.e., the coding region runs from -26.65 to -25.85). Figure 5b shows the result of mapping the second and third coding regions. Pupal RNA was annealed to Pst I-, Ava I-, Bst E II-, or Pvu II-cleaved lambda N2 (indicated pN2). S1-resistant fragments

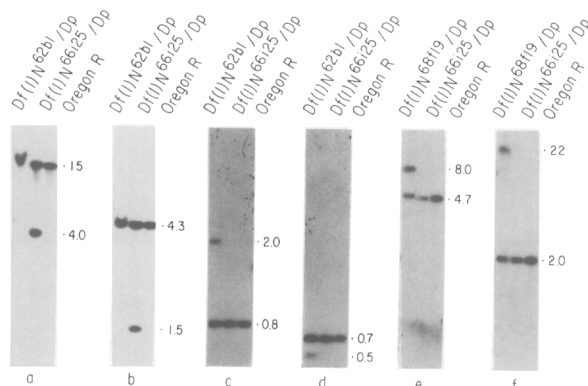


Figure 3. Mapping the Breakpoints in *Notch* of Chromosome Rearrangements

DNA from ten adult flies of the indicated genotype was cleaved with the following restriction enzymes and fractionated through either 0.6% (a, b, e, and f) or 1% (c and d) agarose gels and hybridized with the following probes: (a) Bam HI, (b) Sst I, both hybridized with a 1.9 kb Bgl II/Bam HI fragment (+0.8 to +2.7); (c) Eco RI/Hind III/Pst I, (d) Eco RI/Hind III/Pvu II, both hybridized with a 0.7 kb Pvu II fragment (−19.3 to −18.6); (e) Sst I, (f) Xho I, both hybridized with a subcloned 1.1 kb Eco RI fragment (−25.8 to −24.7). Each of the chromosome rearrangements is maintained in males with *Dp(1:2)51b* as the balancer chromosome. To distinguish between changes in this chromosome and the rearranged X chromosomes, chromosome rearrangements which break at the left end of the genetic map near *N^{55e11}* (*Df(1)N^{62b1}*, c and d, and *Df(1)N⁶⁸¹⁹*, e and f) are compared with *Df(1)N⁶⁶²⁵*, which breaks near *spl* and deletes the left half of the locus (Welshons and Keppy, 1975). The *Df(1)N⁶⁶²⁵* deletion uncovers the duplication in the region including *N^{55e11}* to *spl*. For physically mapping the *Df(1)N⁶⁶²⁵* breakpoint (a and b) this chromosome was compared with *Df(1)N^{62b1}*, which genetically deletes the right half of the *Notch* locus, beginning close to, but to the right of, *N^{55e11}* (Welshons, 1974). Each chromosome rearrangement is also compared with the wild-type stock Oregon R, which in these restriction enzyme digests is identical with the *Dp(1:2)51b* chromosome.

were detected by using the subcloned left terminal Eco RI fragment (−8.4 to −3.5) of lambda N2 as a hybridization probe. Two fragments, of 0.25 and 0.2 kb, are detected in the Pst I and Ava I experiments. Bst E II (−6.35) and Pvu II (−6.20) cleaved the 0.25 kb coding region, producing detectable fragments of 0.17 and 0.2 kb, respectively, the 0.17 kb fragment being resolved on longer exposures. This maps the 0.25 kb coding region to the interval −6.40 to −6.15. The 0.2 kb coding region was not cleaved by any of the enzymes used in Figure 5b. It is detected by a 0.6 kb Pvu II/Pst I fragment (coordinates −6.2 to −5.6) and so lies within this interval to the right of the 0.25 kb coding region. The fourth and fifth coding regions were mapped after Eco RI-, Hind III-, or Ava I-cleaved lambda N2 was annealed to pupal RNA. A ³²P-labeled 1.3 kb Eco RI fragment (coordinates −3.5 to −2.2) of lambda N2 detects a fragment of 0.74 kb in the Ava I experiment (Figure 5c). The 0.74 kb fragment is cleaved by Hind III (−2.30) and Eco RI (−2.20), producing fragments of 0.60 and 0.70 kb, respectively (Figure 5c). This positions the 0.74 kb coding region at coordinates −2.90 to −2.16. A 0.13 kb coding region was detected in the Eco RI and Hind III experiments (Figure 5c). Ava I cleavage of this coding region was confirmed by rehybridizing the filter with

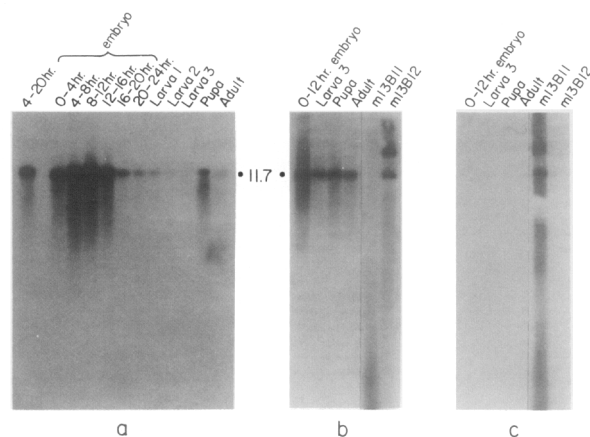


Figure 4. Developmental Profile and Direction of Transcription of the 11.7 kb RNA

(a) 25 μ g of total poly(A)⁺ RNA from the indicated stages was electrophoresed through a 0.6% agarose-formaldehyde gel and hybridized to lambda N2 (−8.4 to +6.2). The size of the RNA was determined using restriction fragments of wild-type lambda and Charon 4 DNA as markers. The major RNA, designated 11.7 kb, has a mobility of 7.8 kb when compared to the DNA size standards. However, this has been multiplied by 1.5 to account for the difference between the known sizes of rRNA and that determined from the DNA markers on these gels. (b) and (c) 10 μ g of total poly(A)⁺ RNA (preparations differ from those shown in a and are more broadly staged) from the indicated stages was fractionated on a 1% agarose-formaldehyde gel alongside approximately 2.5 ng of single-stranded m13B11 and m13B12. These contain a 1.75 kb Bam HI fragment (+2.7 to +4.5) cloned in m13mp9. After transfer to Genescreen, the filter was cut in two, half was hybridized with viral m13B11, (b), and half with m13B12 (c) radiolabeled by the procedure of Hu and Messing (1982). Only m13B11 hybridizes with the *Drosophila* RNA; the Bam HI fragment in this clone is orientated in the RF form such that the Sst I site at +3.2 is closest to the Eco RI site of M13mp9. This only detects transcripts going from left to right of Figure 2. m13B12 has the Bam HI fragment in the opposite orientation; it hybridizes with m13B11 (c) but not with the *Drosophila* RNA.

an Eco RI/Ava I fragment (−3.5 to −2.90; data not shown). This RNA coding region maps to approximately −3.0 to −2.8.

Similar experiments (data not shown) were carried out to localize four other coding regions of the 37 kb transcription unit. The boundaries of these coding regions are coordinates −0.20 to +7.05 for the sixth, 7.15 to 7.52 for the seventh, and 7.60 to 8.07 for the eighth coding region. The ninth (3') coding region is heterogeneous in size, and three sizes have been detected. In all cases this region starts at 8.20 and ends at 8.77, 9.20, or 9.80. Further analysis of coding regions six to nine is presented below. The sum of the coding regions detected by S1 mapping (including the largest 3' coding sequence) is 11.8 kb, which is close to the size of the RNA homologous to this region, determined by gel electrophoresis. To simplify this presentation, the apparent mixture of RNA size classes will be referred to as 11.7 kb.

To show that these coding regions are spliced, the two-dimensional S1 mapping procedure of Favaloro et al. (1980) was used. DNA-RNA hybrids are prepared as in conventional S1 mapping. They are fractionated on a neutral gel, which is then equilibrated in alkali gel buffer to

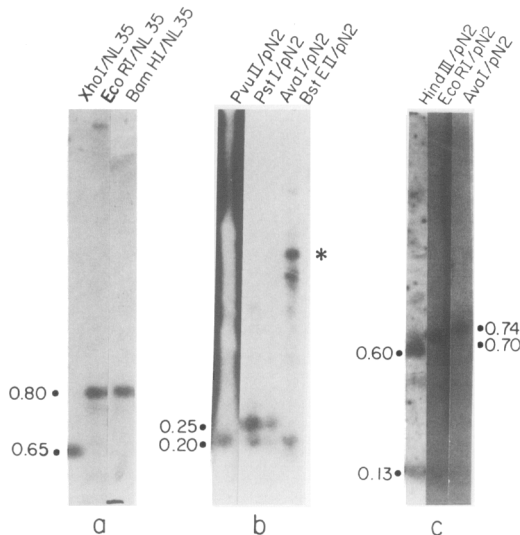


Figure 5. One-Dimensional S1 Mapping of the 11.7 kb RNA

(a) 50 ng of lambda NL35, digested with the indicated restriction enzymes, was annealed with 10 μ g of poly(A)⁺ pupal RNA. After S1 digestion, the hybrids were fractionated on a 1.5% alkali agarose gel and hybridized with an Xho I/Eco RI fragment (–26.5 to –25.8). (b) and (c) 100 ng of lambda N2 (indicated pN2), after digestion with the indicated restriction enzymes, was annealed to 5 μ g of poly(A)⁺ pupal RNA and then digested with S1. The hybrids were fractionated through a 2.5% alkali agarose gel and hybridized with (b) the subcloned left terminal Eco RI fragment (–8.4 to –3.5) or (c) subcloned 1.3 kb Eco RI fragment (–3.5 to –2.2) of lambda N2. Bands marked with an asterisk are due to DNA–DNA reannealing of fragments homologous to the ³²P-labeled hybridization probe.

destroy the RNA part of the hybrid, and electrophoresis is continued after the gel is turned through 90°. If an RNA is unspliced, the DNA it protects from S1 digestion will have the same mobility on neutral and alkali dimensions and will form a spot on the “diagonal.” If it is spliced, the alkali will remove the RNA that held the noncontiguous coding regions together in the neutral dimension. The resulting fragments have a higher mobility in the alkali dimension and produce two or more (depending on the number of intervening sequences in the transcription unit) spots below the diagonal on a “vertical tie line” (Favaloro et al., 1980). Spots on the same vertical tie line result from noncontiguous coding regions being spliced together to form a single RNA. The mobility of the spot on the alkali dimension is the length of the coding region. The sum of the spots on a vertical tie line (if no fragments comigrate) is the length of the transcript (or DNA homologous to the transcript). Alternatively, this length can be obtained from the mobility of the RNA–DNA duplexes in the first dimension.

Initially, experiments were carried out to demonstrate a vertical tie line of coding regions one to six by annealing pupal RNA to N2 and NL35, followed by hybridization with a probe homologous to all these coding regions. These failed because hybridization of the probe to the bimolecular annealing of N2 and the 11.7 kb RNA (labeling coding

regions two to six) obscured any hybridization due to trimolecular annealing. To cope with this, the annealing of N2, NL35, and the 11.7 kb RNA, after electrophoretic fractionation and blotting, was hybridized with an approximately equimolar mixture of Xho I/Eco RI (–26.5 to –25.8) and Xho I/Hind III (–6.8 to –5.4), restriction fragments homologous to coding regions one to three. This is shown in Figure 6A and diagrammatically in Figure 6B. Coding regions two and three are similar in size (0.25 and 0.2 kb) and comigrate as a single spot (a), which is displaced from the diagonal. These coding regions were part of a 7 kb RNA–DNA duplex in the first dimension. This corresponds to the summed lengths of the coding regions of N2. Coding region 1 is 0.8 kb and forms spots b and b’; the former is displaced from the diagonal and was part of an RNA–DNA heteroduplex of about 8 kb in the first dimension. This is the expected result if the 11.7 kb RNA is formed by splicing 0.8 kb of RNA transcribed from NL35 to 7 kb of RNA from N2. Spot a results from bimolecular annealing of the coding regions in N2 to the 11.7 kb RNA, and spot b results from further annealing of this hybrid to NL35 in a trimolecular reaction. Spot b’ results from bimolecular annealing of the 11.7 kb RNA to NL35. If N2 and NL35 were homologous to different 11.7 kb RNAs, then all of the 0.8 kb coding region would be in spot b’, since only 0.8 kb of homology to the 11.7 kb RNA has been found in NL35 (Figure 5a and related text). Because all of the bimolecular annealing of the 11.7 kb RNA and NL35 appears to be resolved as a single, 0.8 kb spot on the diagonal, Figure 6A also confirms that no additional coding regions (>130 bp) homologous to the 11.7 kb RNA are found in NL35.

To show splicing between coding regions six to nine, lambda NR311 was annealed to pupal RNA. This phage contains the 3’ 1.2 kb of coding region six plus the other three coding regions to its right (Figure 2). After transfer to a filter from a two-dimensional gel, the annealing was first hybridized with a 1.5 kb Eco RI fragment (6.8 to 8.3) that was homologous to coding regions six to nine. This is shown in Figure 6C and diagrammatically in Figure 6D. The complexity of the experiment (i.e., the large number of vertical tie lines) results from the size heterogeneity of coding region nine and from “cut through” (Favaloro et al., 1980) by S1 nuclease of the RNA holding noncontiguous segments of DNA together. Spots a, b, c, and d on vertical tie line 1 show the splicing of coding regions six (a), seven (c), and eight (b) to the 1.6 kb (3’) coding region nine (d). Similarly, the spots on the second vertical tie line result from splicing of coding regions six (a’), seven (c’), and eight (b’) to the 1 kb 3’ coding region (e). The smaller 3’ coding region (0.57 kb) comigrates with the eighth (0.47 kb) on this gel, producing a doublet spot b’’f on the third vertical tie line. The remaining spots can be attributed to cut through, although the possibility cannot be excluded that there are transcripts which, for example, lack 3’ coding region sequences altogether. These results were confirmed by hybridizing this filter with two other probes. A 1.9 kb Ava I/Bgl II fragment (+8.2 to +10.1) is specific for

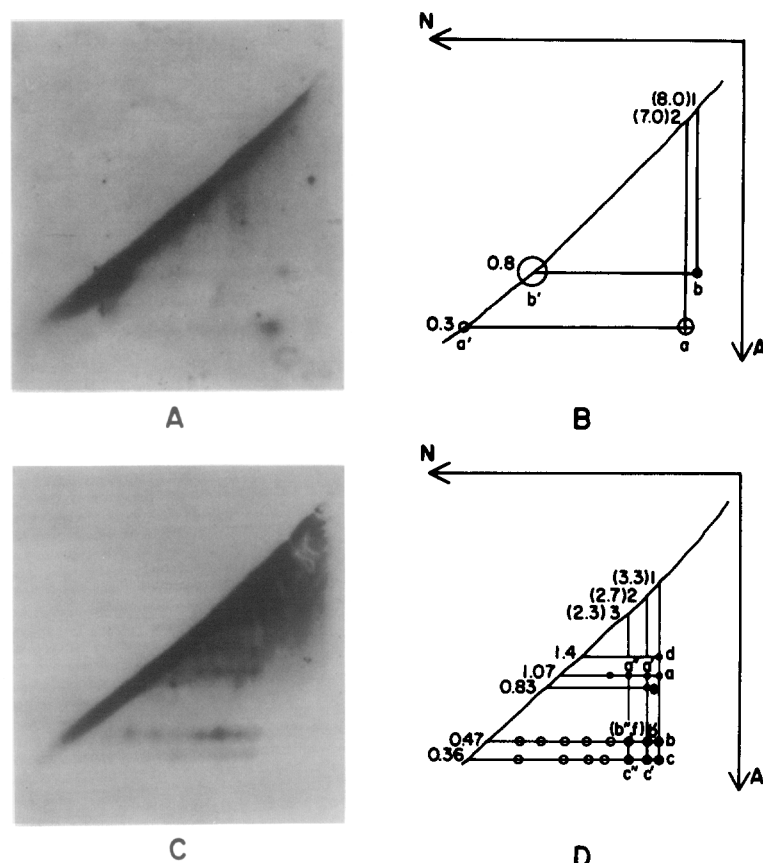


Figure 6. Two-Dimensional S1 Mapping of the 11.7 kb RNA

(A) 100 ng each of lambda NL35 and lambda N2 was annealed with 2–3 μ g of poly(A)⁺ pupal RNA. The S1-resistant hybrids were fractionated on a 0.9% two-dimensional agarose gel with end-labeled Hind III-cleaved lambda cl857 and restriction digests of pAT153 as markers on the second, alkali, dimension. Fractionated DNA was then transferred to a filter. This was hybridized with an approximately equimolar mixture of a Xho I/Eco RI fragment (–26.5 to –25.8) and Xho I/Hind III fragment (–6.8 to –5.4). (B) The result in A, shown diagrammatically with approximate sizes in kilobases. (C) 100 ng of lambda NR311 was annealed with 2–3 μ g of pupal poly(A)⁺ RNA and analyzed on a 1.5% two-dimensional agarose gel with various end-labeled restriction digests of pAT153 as markers. After transfer, the filter was hybridized with a 1.5 kb Eco RI fragment (+6.8 to +8.3). To confirm the identity of the spots, the filter was subsequently hybridized with two additional probes, an Xho I/Hind III fragment (+6 to +7.8) and an Ava I/Bgl II fragment (+8.2 to +10.1) (data not shown). (D) Diagrammatic interpretation of the result in C, with approximate sizes in kilobase pairs.

the 3' coding region, and a 1 kb Xho I/Hind III fragment (+6.0 to +7.8) is homologous to the remaining coding regions (data not shown).

Correlation of Physical and Genetic Maps

Several spontaneously occurring mutations in *Drosophila* are associated with, and in some cases must have been caused by, the insertion of segments of transposable, repetitive DNA (Gehring and Paro, 1980; Bingham and Judd, 1981; Collins and Rubin, 1982; Levis and Rubin, 1982; Rubin et al., 1982; Zachar and Bingham, 1982; Modolell et al., 1983). In order to correlate the genetic map of the *Notch* locus with the transcription map depicted in Figure 2, 24 mutants at the locus were examined for insertions. They were found in seven of the mutants examined. DNA was cloned from three of these *fa*, *fa*^q, and *I(1)N*, and the restriction map of each mutation was compared to that of the wild-type locus (Figure 7). Each is associated with a different copia-like element (see below). The other four mutations, *N*^S, *N*^{55e11}, *N*²⁶⁴⁻⁴⁰, and *N*²⁶⁴⁻¹⁰⁹, were characterized by hybridizing cloned wild-type DNA to restriction digests of total mutant genomic DNA (Southern, 1975). Representative hybridizations are shown in Figure 8, and restriction maps deduced from these experiments are included in Figure 7.

All but two of the mutations, *N*²⁶⁴⁻¹⁰⁹ and *N*^S, have been mapped by recombination to the interval bordered by the breakpoints of *Df(1)N*^{68f19} and *Df(1)N*⁶⁶ⁱ²⁵ (Welshons,

1974; Welshons and Keppy, 1975, 1981). All five of these mutations contain insertions in the same order and relative position as indicated by recombination analyses. Mutants *N*^{55e11}, *fa*, *fa*^q, and *N*^S are of spontaneous origin. *N*^{55e11} genetically defines the distal end of the locus (Figure 1) and cannot be separated by recombination from the breakpoint of *Df(1)N*^{68f19}. However, recombination mapping has placed *N*^{55e11} to the left of the breakpoint of *Df(1)N*^{62b1} (Welshons, 1974). It is associated with a 3.5 kb insertion between the Hind III site at coordinate –26.2 and the Eco RI site at –25.8. This is close to or within the 5' coding region of the 37 kb transcription unit and within the interval in which the *Df(1)N*^{68f19} breakpoint was mapped (–26.2 and –24.7). Further mapping experiments positioned the insertion to the right of an Hpa I site at –26.0 (data not shown).

Recombination analyses have placed the *fa*^q mutation between the breakpoints of *Df(1)N*^{62b1} and *In(1)N*^{76b8} (Welshons and Keppy, 1981). *fa*^q is associated with a 5 kb insertion into the largest intervening sequence of the 37 kb transcription unit (between Hind III sites at –10.0 and –9.2). *fa*^q is homologous to flea (data not shown), which was found 25 kb to the right of the *fa*^q insertion site in the Canton S library. An insertion of at least 8 kb, also within this intervening sequence between Bgl II sites at –11.6 and –10.9, is associated with the *fa* mutation. At least a portion of the *fa* insertion is conserved between members of a repeated sequence family. When the internal

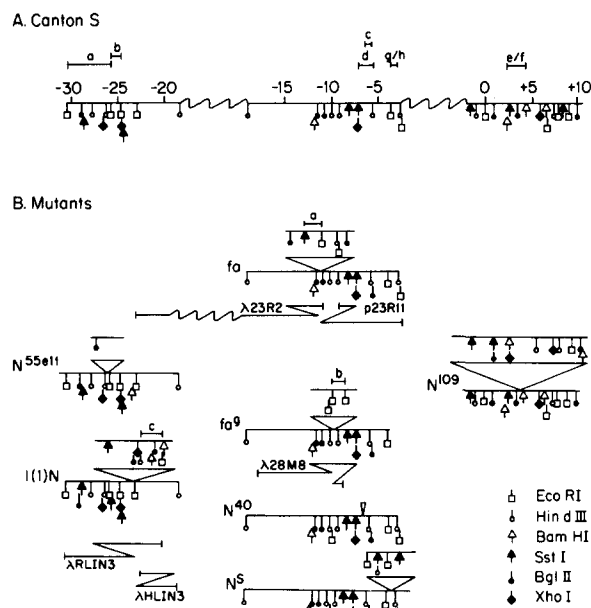


Figure 7. Restriction Maps of Insertion-Associated Mutations of the *Notch* Locus

(A) Restriction maps of the relevant parts of the cloned Canton S DNA with coordinates (in kilobases) shown above them. (B) Restriction maps of insertions associated with *Notch* mutations together with flanking DNA. The cleavage sites for six restriction enzymes are shown, and in most cases only these enzymes were used for localizing the insertions. Two exceptions are N^{264-40} and N^S , which were mapped within a Pst I/Pvu II fragment and an Eco RI/Ava I fragment, respectively (see Figure 2). For construction of clones from mutant DNA see Experimental Procedures. Some restriction site polymorphisms are evident between the flanking DNA of the various mutations and Canton S DNA. N^{264-40} and $N^{264-109}$ are referred to as N^{40} and N^{109} . The size of the $N^{264-109}$ insertion (14.8 kb) is the minimum required to position a single Eco RI cleavage site within it. For the other enzymes only the sites immediately adjacent to the cloned DNA could be positioned. If the insertion were larger (with more Eco RI sites), there would be no change in the sizes of the fragments detected with the cloned DNA. Another source of uncertainty in the size of the $N^{264-109}$ insertion is that it produces quite large fragments difficult to size on 0.6% agarose gels. Lettered bars above the Canton S map (A) and maps of cloned insertions (B) show the locations of probes used in Figures 8 and 9, respectively.

Eco RI/Sst I fragment of the *fa* insertion (Figure 7) is hybridized to genomic DNA cleaved with the same enzymes, a strongly hybridizing fragment that comigrates with that of the cloned DNA is detected (Figure 9a). This is similar to the result obtained when an internal fragment of *fa*⁹ (flea) is hybridized to appropriately cleaved genomic DNA (Figure 9b).

The N^S mutant, supplied by A. Schalet, has not been mapped by recombination, but was found to contain a 5 kb insertion relative to the parental Amherst strain in which it arose, between coordinates -3.5 (Eco RI) and -2.90 (Ava I). This insertion is closely associated with the fourth and fifth coding regions of the transcription unit. The N^{264-40} (N^{40} in Figure 7) mutation maps by recombination between the breakpoints of $Df(1)N^{62b1}$ and $ln(1)N^{76b8}$ (Welshons and Keppy, 1981). It was induced by x-rays and contains a 0.4 kb insertion (Figure 7) between coordinates -6.2 (Pvu II) and -5.6 (Pst I) which is possibly

located in the second or third coding regions of the 37 kb transcription unit (Figure 2).

The origin of the mutant $l(1)N$ is unclear (see Discussion). It is 0.01 map units to the right of N^{55e11} (Welshons, personal communication). DNA from $l(1)N$ contains an 8.5 kb insertion near the left end of the 20 kb intervening sequence (between coordinates -24.2 (Sst I) and -23.0 (Eco RI), about 2.3 kb to the right of the insertion associated with N^{55e11} . An internal restriction fragment from the $l(1)N$ -associated insertion is fairly well conserved within other copies of this repeated element in genomic DNA (Figure 9c). The restriction map of this insertion corresponds to that of the *roo/B104* element described by Meyerowitz and Hogness (1982) and Scherer et al. (1982). The x-ray-induced mutant $N^{264-109}$ lies outside the region defined by $Df(1)N^{68119}$ and $Df(1)N^{66125}$, but has been mapped by recombination 0.003 map units to the right of *spl*. $N^{264-109}$ (N^{109} in Figure 7) contains an insertion of at least 14.8 kb between coordinates $+3.7$ (Bgl II) and $+4.4$ (Bam HI). This is within the sixth coding region of the 11.7 kb RNA.

Discussion

A total of 120 kb of DNA has been cloned from the proximal 3C region of the X chromosome, and the positions of four *Notch* chromosomal rearrangement breakpoints, spanning about 25 kb of DNA, have been mapped within this DNA. To locate the *Notch* locus more precisely, 24 single site mutations were examined for the presence of cytologically invisible chromosome rearrangements. Eleven of these mutations (N^{264-40} , $N^{264-103}$, $N^{264-107}$, $N^{264-109}$, N^{Nic} , N^{60g11} , $l(1)N^2$, $l(1)N^3$, $l(1)N^B$, *spl*, and *fa*^{no}) were induced by x-rays or gamma-rays. The physical maps of only two of these, N^{264-40} and $N^{264-109}$, involve rearrangements, and in both cases these are insertions. None of four chemically induced mutations (*nd*, Ax^{E2} , Ax^{982} , and Ax^{16172}) was associated with detectable changes in the physical maps. Seven of the mutations were spontaneous (N^S , N^{55e11} , N^{Co} , N^{124a} , *fa*, Ax^S , and *fa*⁹). Four of these were associated with insertions. A similar figure of 53% has been found in a larger sample of spontaneous mutations at the *white* locus (Zachar and Bingham, 1982). The origin of two of the mutations, *nd*² and $l(1)N$, is unclear, although the latter was found on a chromosome with the x-ray-induced $N^{264-107}$ mutation (Welshons, personal communication). *nd*² is not associated with a detectable change in the physical map, but $l(1)N$ contains an insertion. The genetic locations of most of the 24 mutations that have been characterized are shown in Figure 10. Those that are not shown in Figure 10, with the exception of N^S , which has not been mapped by recombination, lie to the right of *spl*.

All seven of the mutation-associated insertions described above and all four chromosome rearrangement breakpoints map within a 37 kb transcription unit. Recombination data are available for the four rearrangement breakpoints and for six of the insertion-associated mutants (N^S has not been mapped). The physical locations of

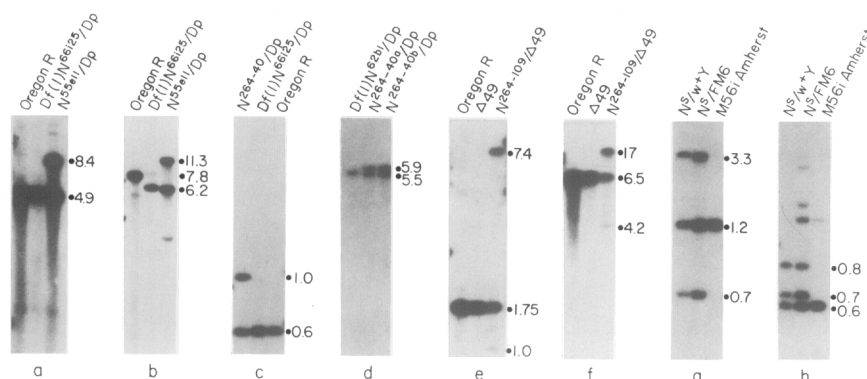


Figure 8. Representative Hybridizations Showing the Presence of Insertions Associated with *Notch* Mutations

Each lane contains DNA from ten flies cleaved with various restriction enzymes and fractionated through either 0.6% (a, b, d, e, and f) or 1% (c, g, and h) agarose gels and transferred to Genescreen. The restriction enzymes and hybridization probes used were (a) Eco RI, 4.9 kb Eco RI fragment (−30.7 to −25.8), (b) Hind III, 1.1 kb Eco RI fragment (pN351.1; −25.8 to −24.7), (c) Pvu II/Pst I, 0.6 kb Pvu II/Pst I fragment (−6.2 to −5.6), (d) Sst I, 1.4 kb Xho I/Hind III fragment (−6.8 to −5.4), (e) Bam HI and (f) Eco RI, both hybridized with a 1.75 kb Bam HI fragment (+2.7 to +4.5), (g) Eco RI/Hind III and (h) Eco RI/Ava I, both hybridized with a 0.6 kb Eco RI/Ava I fragment (−3.5 to −2.90). Each of the filters shown was previously hybridized with several probes to localize the insertions, which accounts for the minor bands present in some lanes as a result of residual amounts of the previous probe. In addition, some bands are due to the pAT153 plasmid vector as they were not observed with gel-purified fragments. The bands above 0.8 kb in (h) result from partial digestion. The recessive lethal mutations at *Notch* are maintained either in males with the autosomal duplication *Dp(1:2)51b* or the Y duplication *w⁺Y*, or in heterozygous females with either the balancer X chromosome, *FM6*, or *ln(1)dl-49* (referred to as $\Delta 49$ in the figure). In most experiments, to distinguish between changes at the mutated *Notch* locus and those at the wild-type locus covering its lethality, DNA from females heterozygous for *Notch* mutations and a balancer are compared to DNA from flies with only the balancer chromosome. When the *Notch* mutations are in males with the autosomal duplication they are compared with males with the same duplication and either *Df(1)N^{62b1}* or *Df(1)N^{66b5}*. The former uncovers the duplication to the right of *N^{55e11}* (Welshons, 1974) and physically to the right of coordinate −18.6. The latter uncovers the duplication to the left of *sp1* (Welshons and Keppy, 1975) and physically to the left of coordinate +0.8. *N^S* is carried in both males and females, and both are compared with the parental Amherst strain to distinguish changes on the mutant chromosomes. In the figure, except for d, g, and h, the mutations are also compared to Oregon R. In the examples, shown, this strain has the same restriction sites as *Dp(1:2)51b*, $\Delta 49$, and Canton S. One exception is that *Dp(1:2)51b* has an additional Hind III site at −20.8 which produces a 6.2 kb Hind III fragment homologous to pN35R1.1 instead of 7.8 kb (see the first two lanes of b). Two examples of *N²⁶⁴⁻⁴⁰* are shown in d; one, *N^{264-40a}*, which is also seen in c, contains on the same chromosome *Df(1)w^{67k30}* and *fa^{mo}* (see Keppy and Welshons, 1977). The positions of each of the above probes are indicated in Figure 7A, where they are named according to the panel they were hybridized with (i.e. a–h) in this figure. The sizes (in kilobase pairs) of DNA fragments are shown beside the lanes of this figure.

insertions associated with the latter six mutations are colinear with the positions of the mutations themselves, as determined by recombination analyses. Each insertion also occurs in an appropriate subinterval of the *Notch* locus, as defined by recombination tests involving the single site mutations and chromosome rearrangement breakpoints (Welshons, 1974; Welshons and Keppy, 1975, 1981). Although these observations do not prove that each of these insertions causes a mutation, the correlations suggest that the six insertions are responsible for the associated mutant phenotypes. Evidence that the insertion found on the *N^S* chromosome generates this mutant phenotype comes from an analysis of the wild-type stock in which it arose. The 5 kb insertion is not found in these parental chromosomes (Figure 8), again suggesting that the mutant phenotype is caused by the insertion.

A factor for converting recombination units at *Notch* into kilobases of DNA can be obtained from the positions of mutations associated with insertions. The distance from *N^{55e11}* to *l(1)N* is 0.01 map units and 2.3 kb, giving 230 kb per map unit; *N^{55e11}* to *fa⁹* is 0.06 units and 16.4 kb, which equals 273 kb per map unit; from *fa⁹* to *N²⁶⁴⁻⁴⁰* is 3.7 kb and 0.013 map units, or 285 kb per map unit. *N²⁶⁴⁻¹⁰⁹* is 0.003 map units from *sp1*, which, in turn, is 0.04 units from *fa⁹*, giving a total of 0.043 units and 13.6 kb. This equals

316 kb per map unit. The mean of all of these numbers is 276 kb per map unit, and this was used to produce the correlated genetic and physical maps of the *Notch* locus shown in Figure 10. As a result, each of the insertions described in this study falls within one or two kilobases of the position predicted from the recombination map.

Most of the mutations at the *Notch* locus that have been localized by recombination are clustered in the proximal 25% of the recombination map. With the exception of *N²⁶⁴⁻¹⁰⁹*, these were not associated with detectable physical changes in the DNA, and the majority of them generate an *N* phenotype. A comparison of the physical and recombination maps in Figure 10 shows that these mutations are apparently in the interval encompassing the largest (sixth) coding region of the 37 kb transcription unit and three tightly linked coding regions to its right.

The insertion in *N²⁶⁴⁻¹⁰⁹* is located in the sixth coding region. It presumably produces an *N* phenotype by introducing new sequences into each of the transcripts from the *Notch* locus. Previous recombination analyses have placed only two *N* mutations outside the four coding regions covered above. These are *N^{55e11}* and *N²⁶⁴⁻⁴⁰*. *N^S*, which has not been mapped by recombination, is also found to the left of these coding regions. *N^{55e11}* carries an insertion closely associated with the 5' coding region. The

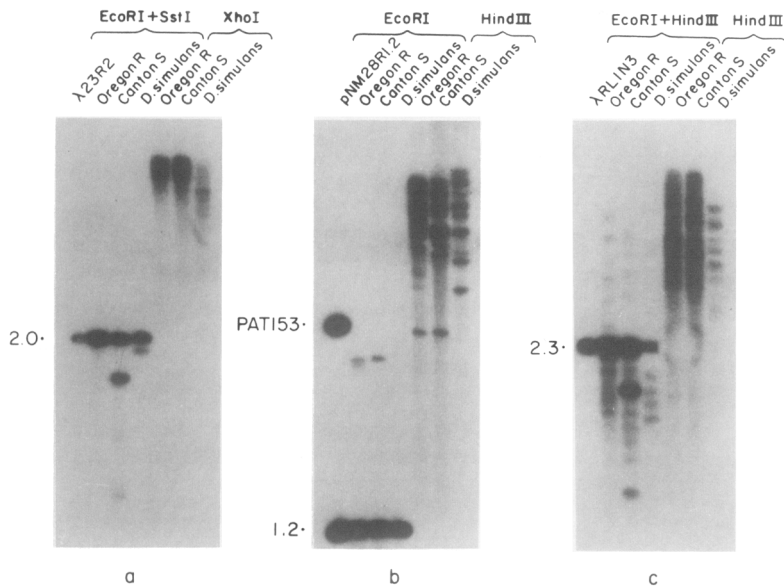


Figure 9. The Mutation-Associated Insertions of *fa*, *fa*^o, and *I(1)N* Are Internally Conserved

Above each lane is shown the source of the DNA and the enzymes with which it was cleaved. The DNA of five flies (approximately 0.3 μ g) was used for the genomic digests; and 0.4 ng, 0.5 ng, and 4 ng of lambda 23R2, pNM28R1.2, and lambda RLIN3, respectively, for the remainder. The DNA was fractionated through 0.6% gels and transferred to Genescreen. The probes are all internal restriction fragments, either gel-purified or subcloned from lambda clones of the mutation-associated insertions. The probes are: (a) 2 kb Eco RI/Sst I fragment of *fa*; (b) 1.2 kb Eco RI fragment (pNM28R1.2) of *fa*^q; (c) 2.3 kb Eco RI/Hind III fragment of *l(1)N*. The locations of these probes are shown in Figure 7B. Probes are labeled in that figure according to the panel they were hybridized with (i.e. a-c) in this figure. When each probe is hybridized to genomic DNA digested with an enzyme that cleaves the cloned insertion once or not at all, multiple fragments are seen that vary in size between strains and species. In contrast, when each probe is hybridized to DNA digested with the same enzymes used to produce the probe, the majority of the hybridization is limited to a fragment comigrating with the internal fragment of the cloned insertion. This is one of the characteristics of the copia-like class of repetitive DNA. For *fa*, *fa*^q, and *l(1)N*, further support for a copia-like structure comes from the finding of homology between fragments containing the termini of the insertion (S. Kidd, unpublished data).

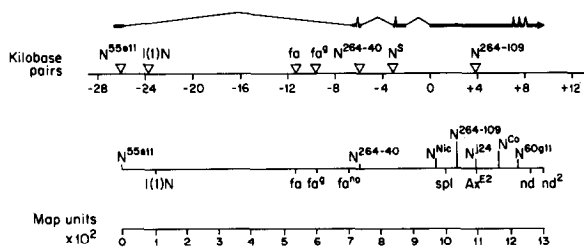


Figure 10. Correlation between Genetic and Physical Maps of the *Notch* Locus

The positions of the RNA coding regions and insertion-associated mutations are shown above a coordinate scale in kilobase pairs. Below is a recombination map of the locus which has been aligned with the physical map assuming 276 kb per map unit. The positions indicated for mutations not associated with detectable changes in the physical map should be regarded as estimates. Inaccuracies of 1 or 2 kb are expected from the analysis of insertion-associated mutations (see Discussion). The 13 divisions represent 0.13 map units, as in Figure 1.

insertion is within coordinates -26.0 and -25.8, and the 3' end of this coding region is at -25.85. Although it is not known whether the insertion falls within this coding region, RNA from *N*^{55e11}/+ embryos contains two smaller *Notch* poly(A)⁺ transcripts as well as the 11.7 kb RNA. All three RNAs hybridize to the Xho I/Eco RI fragment (coordinates -26.5 to -25.8) containing part of the 5' coding

region, and the smaller transcripts have the same polarity as the 11.7 kb RNA (S. Kidd and M. Young, unpublished result). This suggests that insertion at N^{55e11} is associated with premature termination or aberrant processing of *Notch* transcripts. N^{264-40} was mapped within a 0.6 kb restriction fragment (-6.2 to -5.6) that is homologous to the second and third coding regions. N^S is located within a 0.6 kb Eco RI/Ava I fragment (-3.5 to -2.90) that is partially homologous to the fourth coding region. On the basis of these results, it is likely that these four widely separated *N* mutations fail to complement each other and produce the same phenotype because they affect coding regions, or sequences adjoining them, that are spliced together to form functional *Notch* RNA.

The 37 kb transcription unit in which these mutations fall produces transcripts that vary in size. The variation detected is due to the location of the 3' end of each transcript rather than to differential splicing. However, the data presented do not rule out the possibility that differential splicing occurs. Many of the internal RNA coding regions are quite small, and one or more of these might be eliminated in the production of some 11.7 kb RNA. Accordingly, the maps of the transcription unit presented in Figures 2 and 10 represent the structure of the largest RNA that might be produced.

The variable use of 3' coding sequences is possibly similar to dihydrofolate reductase mRNA, where size variation occurs in the 3' untranslated portion of the message; each dihydrofolate reductase mRNA produces an identical protein (Setzer et al., 1980). Figure 10 predicts that *nd* and *nd*², which genetically define the proximal end of the *Notch* locus, lie near the 3' end of the 37 kb transcription unit. These mutants, when homozygous, are phenotypically similar to *N*/+ heterozygotes. Because *nd* and *nd*² could not be physically mapped, their exact locations are uncertain. They may fall within a translated portion of the 37 kb transcription unit, but reduce the activity of its product(s) to a lesser degree than do *N* mutants. Alternatively, they may be located in an untranslated portion of the message and effectively lower the dose of functional *Notch* product(s) by altering the stability or translatability of one or more of the RNA size classes. If *nd* and *nd*² occupy sites much farther to the right (more than 1 kb from the position indicated for *nd*² in Figure 10), they would lie within a completely different transcription unit (S. Kidd, unpublished data).

As reviewed in the Introduction, the recessive mutations, *l(1)N*, *fa*, and *fa*⁹, show some intra-allelic complementation. Each of these mutations was found to be associated with the insertion of a copia-like transposable element into the large intervening sequence. The copia-like elements are often heavily transcribed, and elements from different families can show unrelated patterns of RNA expression during development (Scherer et al., 1981; Schwartz et al., 1982). Although it is not clear how a DNA insertion within the intervening sequence causes a mutant phenotype, the phenotypes of these mutants might be a reflection of stage-specific disruption of *Notch* locus activity through expression of the associated transposable elements. For example, transcripts homologous to the *roo*/B104 element associated with *l(1)N* are found approximately 2.5 hr after egg laying (Scherer et al., 1981), and fall off in abundance within 12 hr. Very little *roo*/B104 RNA is found in larvae, pupae, or adults (S. Kidd, unpublished data). Thus the peak of expression of this element corresponds well with the embryonic lethal period of *l(1)N* (see also Wright, 1970). Such an influence of insertion elements on *Notch* locus activity might also explain some of the observed intra-allelic complementation patterns.

The phenotypes of many of the mutants at *Notch* are temperature dependent (Shellenbarger and Mohler, 1975). Most of these mutations appear to be aberrations in the coding regions that are too small to be detected in these experiments. An exception is the *fa* mutation which is associated with an 8 kb copia-like insertion into the presumably nontranslated large intervening sequence. The response to temperature of *fa* flies could be due to temperature-dependent alterations in the structure or activity of the inserted DNA sequence, or its products, rather than to a modified *Notch* protein.

Most phenotypes associated with mutants at the *Notch* locus appear to be due to defects in the expression or function of the product(s) of the 37 kb transcription unit.

N mutants are associated with reduced activities of four mitochondrial enzymes, and it has been proposed that the *Notch* locus contains the structural genes for these proteins (Thorig et al., 1981). Genetic and developmental analyses have shown that mutations at the *Notch* locus are generally associated with lethality during embryogenesis or with defects at pupation which are seen in the adult (Wright, 1970; Shellenbarger and Mohler, 1975, 1978). RNA produced from the 37 kb transcription unit is developmentally regulated. It is most abundant during embryogenesis and pupation. Whether the mitochondrial enzymes show similar peaks of activity is unknown. In vitro manipulation of cloned *Notch* DNA should allow a direct examination of the protein-coding potential of the locus.

Experimental Procedures

Drosophila Stocks

D. melanogaster stocks were maintained at 25°C on standard cornmeal or grape agar. Mutant *D. melanogaster* strains were obtained from W. Welshons, Iowa State University; A. Schalet, State University of Leiden, The Netherlands; and the Mid-America Drosophila Stock Center, Bowling Green, Ohio. A brief review of the origin of most of the mutations analyzed in this study can be found in Lindsley and Grell (1968). Additional references are cited in the text.

Isolation of Nucleic Acids

DNA from adult flies was prepared as follows. Up to 120 flies were homogenized in 300 μ l of 10 mM Tris-HCl (pH 7.5), 60 mM NaCl, 10 mM EDTA, 0.15 mM spermine, 0.15 mM spermidine, and 5% sucrose. The homogenate was added to 300 μ l of 1.25% sodium dodecylsulfate (SDS), 0.3 M Tris-HCl (pH 9.0), 0.1 M EDTA, 5% sucrose, and 1% diethylpyrocabonate, in a 1.5 ml Eppendorf tube, and incubated at 65°C. After 45 min, 90 μ l of 8 M potassium acetate was added, and the incubation was continued on ice for a further 60 min. The solution was then spun in a microfuge for approximately 1 min, and the supernatant was removed and extracted with 2 volumes of chloroform. Two volumes of ethanol were added, and the precipitated nucleic acids were immediately collected by centrifugation. The nucleic acids were washed once with 70% ethanol, redissolved in 60 μ l of TE₂ (10 mM Tris-HCl, pH 8.0; 2 mM EDTA), and precipitated by the addition of 140 μ l ethanol. After washing with 70% ethanol, the nucleic acids were redissolved at a concentration of ten flies' worth of DNA per 20 μ l of TE₂. When the DNA was to be cloned, 200 μ g/ml of proteinase K was used in place of diethylpyrocabonate, and the solution was incubated for 2 hr at 37°C. Sarkosyl was then added to 2%, the volume was measured, and 1.16 g/ml of CsCl and 0.05 volumes of 5 mg/ml ethidium bromide were added. The DNA was banded in either a vertical or a fixed angle rotor and collected under ultraviolet light. DNA was extracted with isoamyl-alcohol to remove ethidium bromide and dialyzed against TE₂ to remove CsCl. DNA was concentrated by extraction with isobutanol, followed by dialysis against TE₂. In some cases, the procedure of Schwartz et al. (1982) was followed.

Large-scale (>500 ml) preparations of plasmid DNA were carried out by the modified alkali-SDS procedure of Ish-Horowitz and Burke (1981). For small-scale (<1.5 ml) preparations of plasmid and M13 RF recombinants, the rapid-boiling method of Holmes and Quigley (1981) was followed. Bacteriophage lambda DNA was isolated by phenol extraction of phage particles prepared from plate or liquid lysates by banding on CsCl equilibrium gradients. Small-scale preparations were made by the procedure of Davis et al. (1980).

Total RNAs from the various developmental stages of *Drosophila* were isolated using guanidium isothiocyanate and centrifugation through CsCl cushions as described by Schwartz et al. (1982). Poly(A)⁺ RNA was selected on oligo(dT)cellulose.

Cloning

Phages containing wild-type *Drosophila melanogaster* DNA were isolated from the library of Maniatis et al. (1978). This consists of randomly sheared

Canton S DNA cloned into Charon 4 by Eco RI linkers. Phage libraries from various *Notch* mutants were constructed in lambda gt-lambda B (Thomas et al., 1974), Charon 4 (Blattner et al., 1977), or Charon 30 (Rimm et al., 1980). When cloning into the Eco RI or Bam HI sites of the Charon phages, the arms were separated from the internal fragments by preparative agarose gel electrophoresis. For dephosphorylating 5' termini of restriction fragments prior to cloning, the restriction enzyme was inactivated by heating to 70°C for 10 min, and an excess of calf intestine alkaline phosphatase (CIAP; Boehringer Mannheim) was added and incubated for 60 min at 37°C. It was inactivated either by heating to 70°C for 15 min (for subcloning into plasmids) or by phenol extraction (for constructing genomic libraries). Ligations of *Drosophila* DNA to phage vectors were carried out at a concentration of 250–300 µg/ml with an excess of T4 ligase (New England Biolabs). The DNA was packaged in vitro by the procedure of Scalenghe et al. (1981) except that putrescine was omitted. Efficiencies of up to 4×10^6 pfu/µg of insert were obtained. The phage particles were used to infect *E. coli* KH802 and were screened by plaque hybridization essentially as described by Maniatis et al. (1978).

A phage that contained the *Sgs4* locus at 3C11 was obtained from D. S. Hogness. Several phages were isolated distal to *Sgs4* in order to reach the proximal breakpoint of *ln(1)N^{700b}*. Eco RI fragments from flies containing the inversion together with an autosomal duplication of *Notch* were cloned into Eco RI-cleaved lambda gt-lambda B. They were screened using a 6.9 kb Eco RI fragment just distal to *Sgs4* (see Figure 5 of McGinnis et al., 1980). Small-scale preparations of lambda DNA were made from the hybridizing phages to identify those with the *ln(1)N^{700b}* breakpoint. A restriction fragment from one of these was used to isolate a part of the *Notch* locus from the Canton S DNA library. The rest of the locus was isolated by using terminal fragments of previously cloned phages to isolate overlapping segments of DNA. To minimize the overlap of adjacent phages, a counter-screen using a nearby internal fragment of the previously isolated phage was employed.

Eco RI fragments of the *fa* mutation were originally cloned into Charon 4 to form lambda 23R2 and 23R11. p2311 was subcloned from the latter phage using pAT153 (see below). The *fa*^a insertion was isolated from a partial Mbo I library of *fa*^a DNA cloned in Charon 30. The probe used to isolate *fa* and *fa*^a clones was the 14 kb Eco RI fragment of lambda NL22 (–22.2 to –8.2). The clone lambda RLIN3 was isolated by partially Eco RI-digesting DNA from flies with *ln(1)N* together with a *Notch* duplication, *Dp(1;2)51b*, treating the fragments with CIAP, and ligating them to Charon 4. The probe used to isolate the clone was the 1.7 kb Eco RI fragment (–24.7 to –23.0) of lambda NL35. The clone lambda HLIN3 was isolated by ligating Hind III fragments from the same flies into Charon 30 that had been cleaved with Hind III and then dephosphorylated with CIAP. The phages that were picked hybridized to the right terminal Eco RI fragment (–23.0 to 19.5) of lambda NL35 and the right-hand Eco RI/Hind III fragment of lambda RLIN3 (probe c of Figure 7B).

Restriction fragments were subcloned into pAT153 (Twigg and Sherratt, 1980), or M13mp9RF, or M13mp8RF (Messing and Vieira, 1982) that had been cleaved with the appropriate restriction endonuclease and then treated with CIAP. Recombinant pAT153 was used to transform competent *E. coli* HB101 prepared by the procedure of Morrison (1979), and stored at –70°C. M13mp9 recombinants were used to transform either *E. coli* JM103 or TB1, a *recA* derivative of JM101 constructed by A. Lamond.

S1 Mapping

RNA–DNA duplexes were formed and digested with S1, and subsequently fractionated on one- or two-dimensional gels as described by Favaloro et al. (1980). Details of each experiment are given in the legends to Figures 5 and 6.

Gel Electrophoresis, Transfer to Filters, and Restriction Mapping

Restriction enzymes were obtained from New England Biolabs and Bethesda Research Laboratories. DNA digests were fractionated through Tris borate-buffered gels. Prior to filter transfer (Southern, 1975), the DNA was dephosphorylated in situ in 0.25 M HCl for 30 min (Wahl et al., 1979). After denaturation and neutralization the gels were equilibrated in 25 mM NaPO₄, and the DNA was transferred to Genescreen (New England Nuclear). Alkali gels were neutralized by four 15 min washes in 25 mM sodium phosphate (pH 7.0), then transferred to Genescreen. Low melting point (LMP) agarose gels were run in Tris–acetate buffer. DNA was extracted from LMP agarose

as previously described (Kidd and Glover, 1981). Restriction maps were constructed by single and double digestions of phages. Some fragments were isolated in LMP agarose and digested in situ with a second enzyme. In some cases, when determining the order of small restriction fragments, the procedure of Smith and Birnstiel (1976) was followed except that the partial digestion was carried out in situ in LMP agarose after addition of 1 µg of lambda DNA.

RNA was fractionated through 2.2 M formaldehyde–agarose gels in 10 mM NaPO₄ (pH 7.4) (Rave et al., 1979) and then transferred to Genescreen.

In Vitro Labeling of DNA and Hybridizations

Double-stranded DNA was labeled in vitro by nick translation. A 10 µl reaction in 50 mM Tris-HCl (pH 7.5), 10 mM MgCl₂, 10 mM 2-mercaptoethanol, 50 µg/ml bovine serum albumin, and 25 µM unlabeled dNTPs, contained 25 ng of DNA, 10 pg of DNAase I (Sigma), 4 units of DNA polymerase I (New England Nuclear), and 50 µCi of ³²P-labeled dCTP (Amersham >3000 Ci/mmol). After incubation at 15°C for 30 min, the reaction was stopped by the addition of an equal volume of 1 mg/ml proteinase K, 25 mM EDTA (pH 8.0), and 1% SDS, and then incubated at 37°C for 60 min. It was then made 0.4 M NaOH, incubated at 65°C for 10 min, and spun through a G-50 M column equilibrated in 30 mM NaOH/1 mM EDTA. The specific activity of the DNA was approximately 0.3 to 3 × 10⁶ cpm/µg. It was added directly to the hybridization solution without boiling. Single-stranded DNA probes were prepared using M13 (Hu and Messing, 1982). The primer was from New England Biolabs, and the large fragment of DNA polymerase I from New England Nuclear. Restriction fragments were 3'-end-labeled with reverse transcriptase (Wahl et al., 1979).

Filters were preincubated for 2 hr and, after the addition of ³²P-DNA, hybridized for 16 hr in 5× SSC, 50% formamide, 10× PM (Denhardt, 1966), 500 µg/ml denatured sonicated salmon sperm DNA, 0.4% SDS, 0.1% sodium pyrophosphate, and 10% dextran sulfate (Wahl et al., 1979). The filters were then washed three times for 5 min each time in 0.2× SSC, 0.2% SDS, 0.1% Na-pyrophosphate. After which, DNA filters were washed once at 60°–65°C for 30 min in 0.2× SSC and 0.2% SDS, then twice for 20 min each time at room temperature in 3 mM Tris base. RNA filters were washed twice for 30 min each time in 0.2× SSC, 0.2% SDS, at 60°–65°C. When single-stranded probes were used, the SSC concentration was increased to 0.5×. The filters were dried and autoradiographed using Kodak XAR-5 film with intensifying screens at –70°C. Filters were reused by the procedure of Thomas (1980).

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